🧠 1. SYSTEM OVERVIEW (BIG PICTURE)

You are building:

- A **single-store ecommerce system**
- Mobile-first (very important)
- Mobile money checkout
- Clean, fast, simple UX

👤 2. USER JOURNEY (END-TO-END)

🟢 PHASE 1 — LANDING

User visits homepage:

- Sees featured products
- Can search products
- Can browse categories

🟢 PHASE 2 — PRODUCT DISCOVERY

User can:

- Click product
- View:
    - Images
    - Price
    - Description
    - Stock availability

🟢 PHASE 3 — ADD TO CART

Backend Logic:

- Check stock
- Add item to cart

If:

- Logged in → store in DB (`user_id`)
- Guest → store via `session_id`

🟢 PHASE 4 — CART VIEW

User sees:

- Product list
- Quantity controls
- Total price

Logic:

- Update quantity
- Remove items
- Recalculate total

🟢 PHASE 5 — CHECKOUT

User fills:

- Full name
- Phone number
- Address
- City
- Region

⚙️ 3. CORE CHECKOUT LOGIC

Step 1 — Validate Input

- Required fields
- Phone format (important for MoMo)

Step 2 — Fetch Cart

```
Cart::with('items.product')
```

Step 3 — Calculate Total

```
$total += item.price * quantity
```

Step 4 — Create Order

```
orders:
- status = pending
- payment_status = unpaid
```

Step 5 — Create Order Items

Each cart item → order_items

Step 6 — Save Address

Store in `order_addresses`

Step 7 — Create Payment Record

```
payments:
- status = pending
```

Step 8 — Trigger Mobile Money

- Send request to API (Paystack / Hubtel)
- User approves on phone

💳 4. PAYMENT FLOW (CRITICAL)

📲 What happens:

1. User receives MoMo prompt
2. User enters PIN
3. Payment provider calls your system (callback)

🔁 Callback Logic

If SUCCESS:

- `payments.status = success`
- `orders.payment_status = paid`
- `orders.status = paid`

Then:

🔥 Reduce Stock

```
product.stock_quantity -= quantity
```

Add tracking:

```
OrderTracking:
"Payment received"
```

❌ If FAILED:

- `payments.status = failed`
- `orders.payment_status = failed`

📦 5. ORDER PROCESSING FLOW

After payment:

🟡 Stage 1 — Paid

- Order confirmed

🟡 Stage 2 — Processing

Admin prepares order

🟡 Stage 3 — Shipped

- Dispatch started

🟡 Stage 4 — Delivered

- Customer receives item

🔄 Tracking Table Updates

Each stage inserts:

```
"Order is being processed"
"Order shipped"
"Order delivered"
```

👨‍💼 6. ADMIN SYSTEM FLOW

Admin can:

📦 Manage Products

- Create
- Update
- Delete
- Upload images

📦 Manage Orders

- View orders
- Change status
- Add tracking updates

💳 View Payments

- Confirm payments
- Monitor failures

👥 Manage Users

- Suspend / activate users

🧩 7. SYSTEM COMPONENT FLOW (TECH VIEW)

🛒 CART FLOW

```
Add → Store in DB/session → Update → Checkout
```

📦 ORDER FLOW

```
Cart → Order → Payment → Tracking → Delivery
```

💳 PAYMENT FLOW

```
Pending → Success → Update Order → Reduce Stock
```

🔐 8. IMPORTANT LOGIC RULES

❗ Rule 1 — Stock Safety

- Do NOT reduce stock on add to cart
- Reduce ONLY after payment success

❗ Rule 2 — Idempotency (very important)

Payment callback should not run twice:

```
if ($payment->status === 'success') return;
```

❗ Rule 3 — Cart Cleanup

After successful payment:

```
delete cart items
```

❗ Rule 4 — Order is source of truth

Never depend on cart after checkout

🚀 9. ROUTES STRUCTURE

```
GET   /              → Home
GET   /products      → Shop
GET   /product/{slug}

POST  /cart/add
POST  /cart/update
POST  /cart/remove

GET   /checkout
POST  /checkout

POST  /payment/callback

GET   /order/{id}
```

🎨 10. FRONTEND FLOW (IMPORTANT FOR UX)

Pages:

🏠 Home

- Featured products

🛍️ Product Page

- Add to cart button (sticky on mobile)

🛒 Cart Page

- Simple list
- Big checkout button

💳 Checkout Page

- ONE page only (important)

📦 Order Success Page

- Show:
  - Order number
  - Status
  - Tracking info

🔥 11. WHAT MAKES THIS SYSTEM GOOD

- Simple checkout (no friction)
- Mobile-first
- Clean backend logic
- Proper data separation
- Scalable later

🧠 FINAL FLOW (COMPLETE)

```
User visits site
→ Browses products
→ Adds to cart
```

→ Proceeds to checkout
→ Enters details
→ System creates order
→ Payment initiated
→ User confirms payment
→ Callback updates system
→ Stock reduced
→ Order processed
→ Order delivered